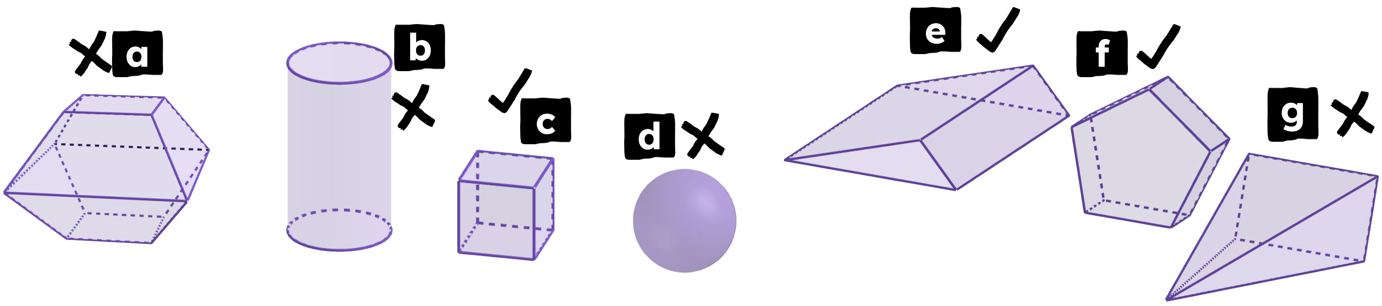
Properties of prisms

Task A: Defining a prism

1) Which of these match the properties of a **prism**?



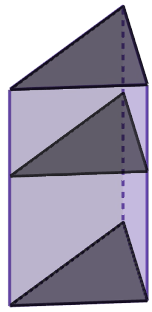
2) Complete the sentences with the missing words provided to give an explanation of what a **prism** is.

A **prism** is a __3D__ object.

All of its 2D **cross sections** must be __polygons__.

In at least one direction, all __parallel__ **cross sections** must

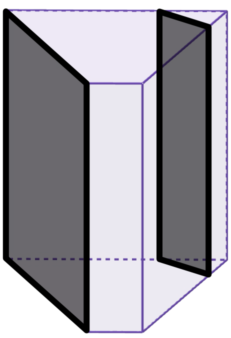
be __congruent__ to each other.



2D | circular | perpendicular | equidistant

3) These diagrams all show the same **prism**.
Which of the diagrams show the correct **cross sections** that help demonstrate it is a **prism**? Explain how you know.

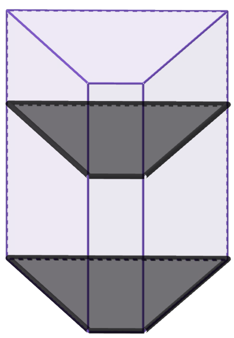
a



These cross sections:
are polygons
are parallel to each other
are not congruent

Therefore, these cross sections do not help demonstrate this is a prism.

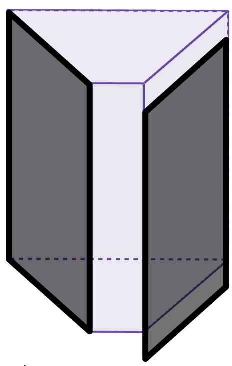
b ✓



These cross sections:
are polygons
are parallel to each other
are congruent

Therefore, these cross sections help demonstrate this is a prism.

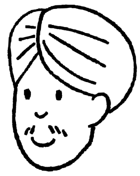
c

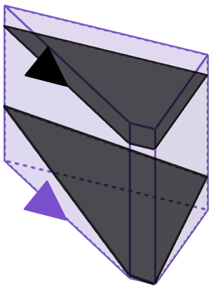


These cross sections:
are polygons
are not parallel to each other
are congruent

Therefore, these cross sections do not help demonstrate this is a prism.

4) Andeep says this 3D shape is not a **prism**, because the two **cross sections** he found are not congruent to each other. Explain why Andeep is incorrect.

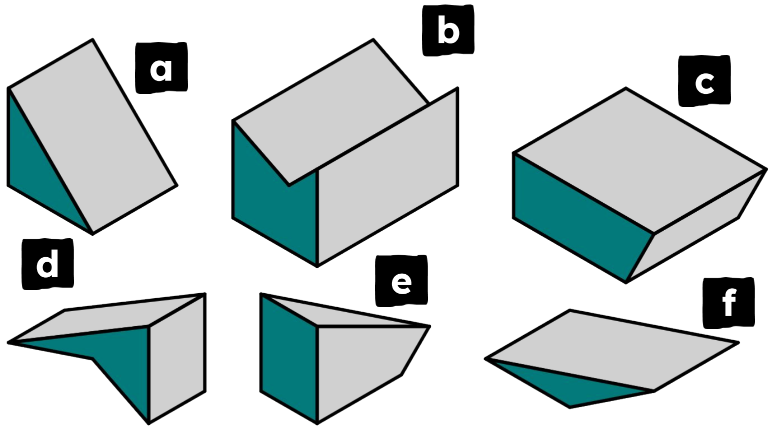




*This might still be a **prism**.*
*Andeep has not taken two **cross sections** that are parallel to each other.*
*One **cross section** looks parallel to the top face of the object, whilst the other **cross section** is at an angle compared to the bottom face.*

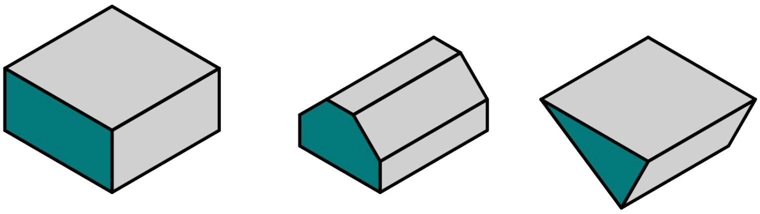
Task B: Categorising prisms

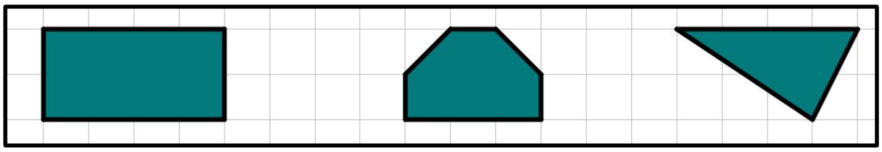
1) Place these 3D shapes into the correct row in the table.



triangular **prism**
a f
quadrilateral **prism**
c d
pentagonal **prism**
b
not a **prism**
e

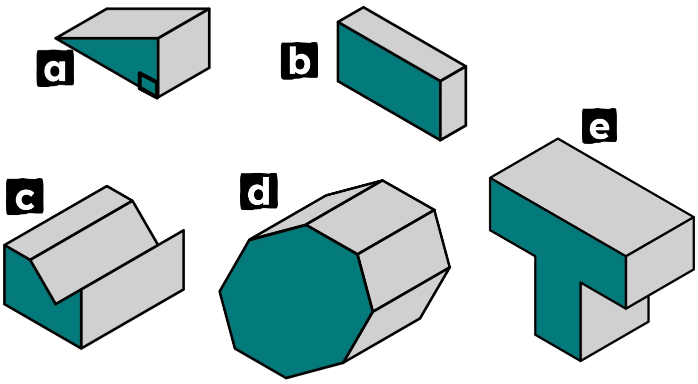
2) On squared paper, sketch the **cross sectional polygon** for each of these **prisms**.





These are only approximate shapes - if your sketch differs slightly, that is okay.

3) Write down a suitable name for each **prism**.



If you came up with a different answer, can you justify it?

a *triangular **prism***
*right-angled triangular **prism***

b *quadrilateral **prism***
*rectangular **prism***

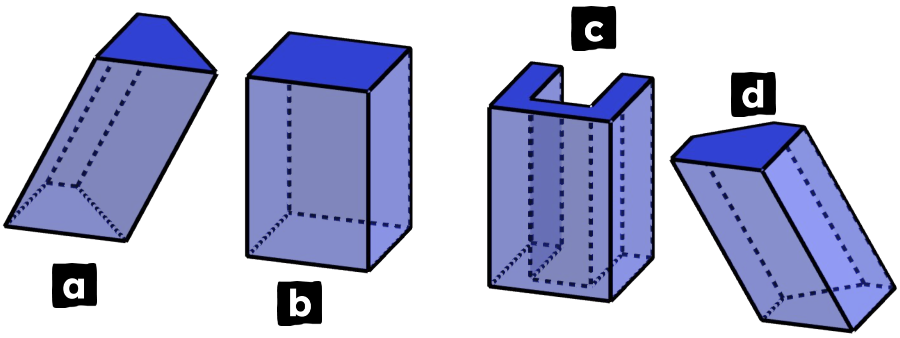
c *hexagonal **prism***
*irregular hexagonal **prism***

d *octagonal **prism***
*regular octagonal **prism***

e *octagonal **prism***
*irregular octagonal **prism***

Task C: Understanding oblique prisms

1) Place these 3D shapes into the correct row in the table.



right **prism**
b c
oblique **prism**
a d

2) For each **prism**, write down a suitable name.

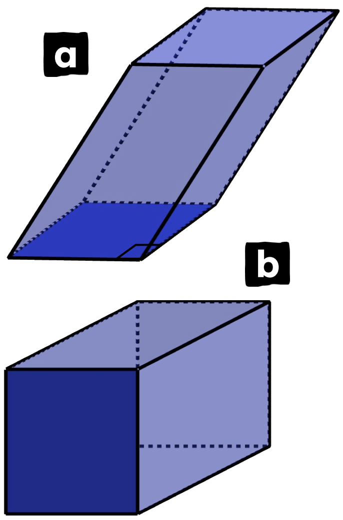
a *oblique trapezium **prism** or oblique quadrilateral **prism***

b *right rectangular **prism** or right quadrilateral **prism***

c *right irregular octagonal **prism***

d *oblique irregular pentagonal **prism***

3) **Prisms** a and b are the same **prism** seen from different perspectives.
Who is correct: Aisha, Lucas, both, or neither?



Aisha: This **prism** is an oblique **prism**.
This is because the rectangles on the top and bottom of **prism** a are parallel to each other, but the other faces are not perpendicular to them.

Lucas: This **prism** is a right **prism**.
This is because the parallelograms on the top and bottom of **prism** b are parallel to each other, and the other faces are perpendicular to them.

Both are correct.

*This can be true for some quadrilateral **prisms**.*

*When looked at in one direction, they are a right parallelogram **prism** (such as in b).*

*When looked at in a different direction, they are an oblique rectangular **prism** (such as in a).*